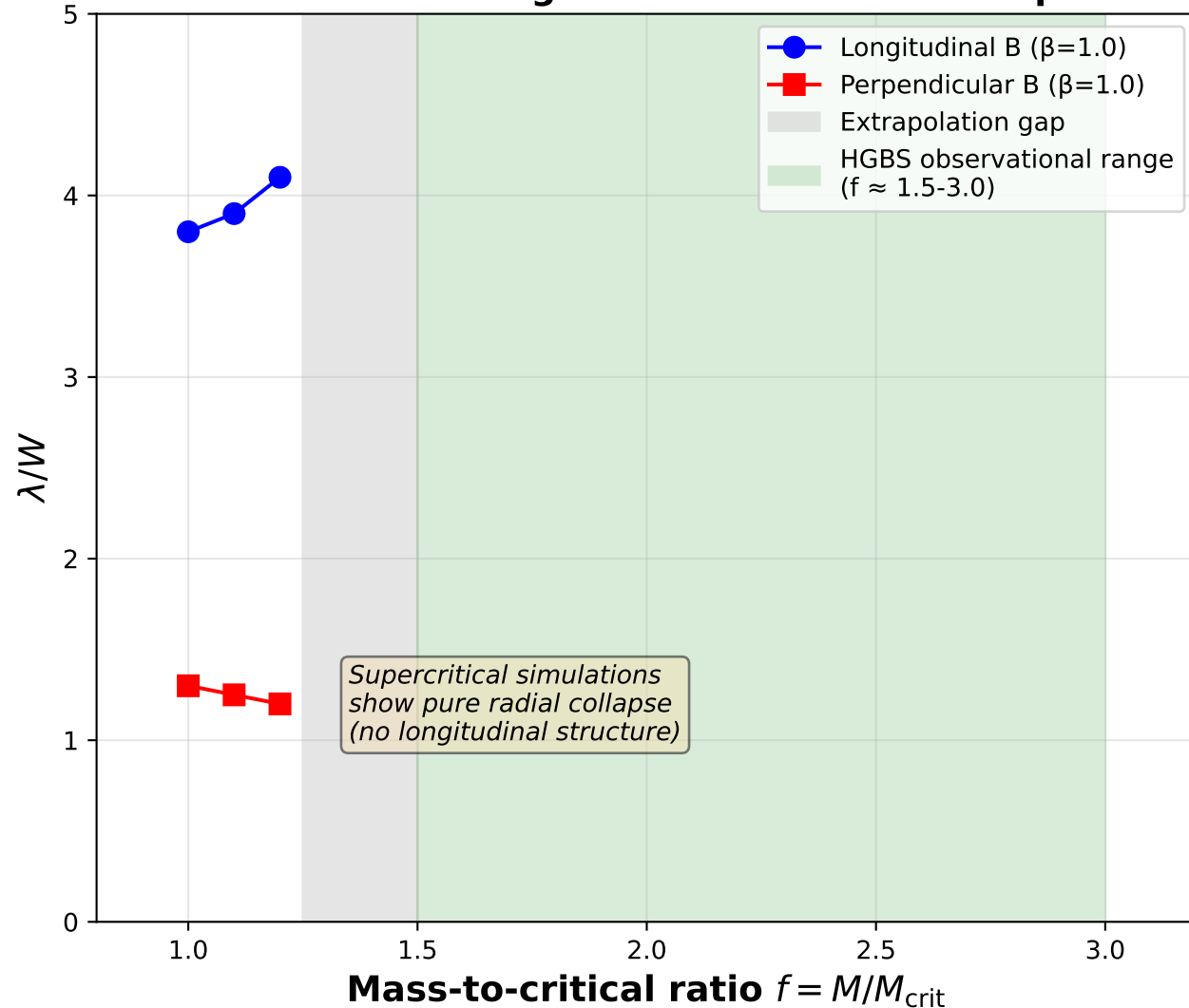


Simulation Coverage vs. HGBS Parameter Space



Measurement vs. Inference: The Fundamental Limitation

Near-critical simulations
($f = 1.0-1.2$)

$\lambda/W = 1.25$ (perpendicular)
 $\lambda/W = 3.8-4.1$ (longitudinal)

Direct measurement

Extrapolation gap
($f = 1.2-1.5$)

NO SIMULATIONS
(design gap)

Extrapolation
(required)

Supercritical simulations
($f = 1.5-3.0$)

NO FRAGMENTATION
(pure radial collapse)

Extrapolation
(required)

HGBS observations
($f \approx 1.5-3.0$)

NN: $\lambda/W = 1.73-2.06$
PM: $\lambda/W = 2.79$

Comparison with theory
(extrapolated)

KEY POINT: Theoretical predictions for HGBS regions rely on extrapolation from near-critical simulations, but the smooth $\lambda/W(f)$ relationship near $f = 1.0-1.2$ does not validate extrapolation to the supercritical regime where radial collapse dominates.